A-Level Physics: Synoptic Question on Seismometer Design

Synoptic Question: The Physics of an Electromagnetic Seismometer

A seismometer is a device used to detect and measure ground motion during earthquakes. A common design involves a heavy "inertial mass" suspended by a spring from a frame attached to the ground. A coil of wire is attached to the mass, which moves within a radial magnetic field produced by a permanent magnet fixed to the frame.

Part A: Oscillations and Resonance

The suspended mass m is 0.50 kg. The spring constant k is 80 N m^{-1}.

1. Define *Simple Harmonic Motion*.
2. Calculate the natural frequency f_0 of the seismometer's mass-spring system.
3. In a real seismometer, the system is often "critically damped."
   - Explain the meaning of critical damping in this context.
   - Sketch a displacement-time graph for the mass if it is displaced and then released in a critically damped system.

Part B: Electromagnetic Induction (EMI)

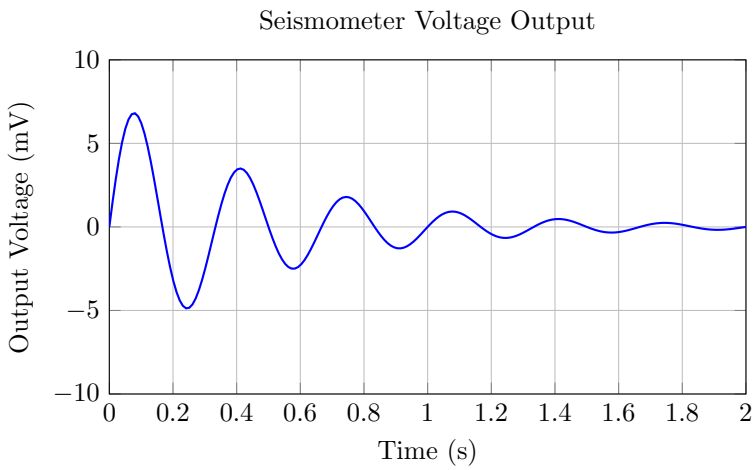
As the ground moves, the frame moves with it, while the inertial mass tends to remain stationary. This causes relative motion between the coil and the permanent magnet.

4. The coil consists of 500 turns of wire, each with a mean radius of 1.5 cm. The radial magnetic field strength B is 0.40 T. Show that the electromotive force (emf) ε induced in the coil is related to the relative velocity v by:
$$\varepsilon = BLv$$
where L is the total length of the wire in the coil. Calculate the value of ε when the relative velocity is 0.12 m s^{-1}.
5. Explain, using Lenz's Law, how the induced current in the coil provides a damping force on the mass.

Part C: Wave Analysis

An earthquake produces P-waves (longitudinal) and S-waves (transverse). A seismometer at a station detects a P-wave at 12:00:00 and an S-wave at 12:00:45.

6. The speed of P-waves is 8.0 km s^{-1} and the speed of S-waves is 4.5 km s^{-1}. Calculate the distance from the seismic station to the epicenter of the earthquake.
7. The graph below shows the voltage output of the seismometer as a function of time for a specific wave pulse.



Using the graph, estimate the frequency of the detected seismic wave and determine the maximum relative velocity of the coil during the first oscillation.

Suggested Answers and Mark Scheme

Part A: Oscillations and Resonance

1. **(2 marks)** SHM is motion where the acceleration is directly proportional to the displacement from a fixed equilibrium position and is always directed towards that position ($a \propto -x$).

2. **(2 marks)** $T = 2\pi\sqrt{m/k} = 2\pi\sqrt{0.5/80} \approx 0.497$ s.
   $f_0 = 1/T = 1/0.497 \approx 2.01$ Hz.

3. **(2 marks)** Critical damping is the condition where the system returns to equilibrium in the shortest possible time without overshooting. Graph: A curve that decays to zero displacement without crossing the x-axis.

Part B: Electromagnetic Induction

4. **(3 marks)** Total length $L = N \times 2\pi r = 500 \times 2\pi \times 0.015 \approx 47.12$ m.
   $\varepsilon = N\frac{\Delta\Phi}{\Delta t} = N\frac{B\cdot\Delta A}{\Delta t} = N \cdot B \cdot (2\pi r) \cdot v = BLv$.
   $\varepsilon = 0.40 \times 47.12 \times 0.12 \approx 2.26$ V.

5. **(2 marks)** Lenz's Law states the induced emf/current acts in a direction to oppose the change that produced it. The current in the coil creates its own magnetic field that exerts a force opposing the relative motion of the coil, effectively acting as a resistive (damping) force.

Part C: Wave Analysis

6. **(3 marks)** Let distance be d. $\Delta t = \frac{d}{v_s} - \frac{d}{v_p}$.
   $45 = \frac{d}{4.5} - \frac{d}{8.0} = d\left(\frac{8.0-4.5}{36}\right) = d\left(\frac{3.5}{36}\right)$.
   $d = \frac{45\times 36}{3.5} \approx 463$ km.

7. **(3 marks)** Frequency: One full cycle occurs in approx 0.33 s. $f \approx 1/0.33 \approx 3.0$ Hz.
   Max Voltage $V_{max} \approx 8$ mV $= 0.008$ V.
   $V = BLv \Rightarrow v_{max} = \frac{0.008}{0.40\times 47.12} \approx 4.24 \times 10^{-4}$ m s^{-1}.

A Level Physics (9702) – Paper 4 Style

Variant 2 (Higher Difficulty): Seismometer Physics

Instructions (CIE P4 style)

- Answer all questions.
- Working must be shown.
- Give answers to an appropriate number of significant figures.
- The total number of marks for this variant is **35**.

Question 1: Dynamic response of an electromagnetic seismometer (20 marks)

A vertical seismometer is modelled as a mass–spring–damper system. The frame is fixed to the ground and oscillates with displacement

$$x_g = X_g \sin(\omega t),$$

where $X_g = 1.8 \times 10^{-3}\,\mathrm{m}$ and $\omega = 2\pi f$.

The inertial mass has $m = 0.80\,\mathrm{kg}$ and is connected to a spring of stiffness $k = 120\,\mathrm{N\,m^{-1}}$. The electromagnetic damping force is proportional to relative velocity, with damping coefficient $c = 5.5\,\mathrm{N\,s\,m^{-1}}$.

The coil has $N = 900$ turns, mean radius $r = 1.2\,\mathrm{cm}$, and moves through a radial magnetic field of flux density $B = 0.50\,\mathrm{T}$.

1.1. Show that the undamped natural angular frequency is approximately $12.2\,\mathrm{rad\,s^{-1}}$. Hence determine the natural frequency in Hz. [3]

1.2. For the damped system, calculate the critical damping coefficient c_c and hence the damping ratio ζ. State whether the system is underdamped, critically damped or overdamped. [4]

1.3. The relative displacement $y = x_m - x_g$ has steady-state amplitude

$$Y = \frac{m\omega^2 X_g}{\sqrt{(k - m\omega^2)^2 + (c\omega)^2}}.$$

Calculate Y when $f = 1.80\,\mathrm{Hz}$. [4]

1.4. Show that the total wire length in the coil is $L = N(2\pi r)$, and determine L. [2]

1.5. The induced emf magnitude is $\varepsilon = BLv_{\mathrm{rel}}$, where $v_{\mathrm{rel,max}} = \omega Y$. Determine the maximum induced emf at $f = 1.80\,\mathrm{Hz}$. [3]

1.6. **Open-ended conceptual task.** During a nearby earthquake, significant signal energy appears in the range $2.5\,\mathrm{Hz}$ to $6.0\,\mathrm{Hz}$. Explain and discuss **two design modifications** to improve the reliability of recorded waveforms for this event.

In your answer, include:

- the physical principle for each modification,

- expected impact on sensitivity / bandwidth / distortion,
- one practical trade-off for each.

[4]

Question 2: Seismic-wave interpretation and uncertainty (15 marks)

A station records first arrivals of P and S waves from a single earthquake. Measured arrival times are:

$$t_P = 14{:}32{:}18, \qquad t_S = 14{:}33{:}08.$$

Assume wave speeds in the crust are:

$$v_P = 7.6\,\mathrm{km\,s^{-1}}, \qquad v_S = 4.3\,\mathrm{km\,s^{-1}}.$$

For this station, timing uncertainty is $\pm 1.0\,\mathrm{s}$ on each arrival pick.

2.1. Determine the distance from the station to the epicentre. [3]

2.2. Using worst-case timing uncertainty, estimate the uncertainty in this distance. [4]

2.3. The station output is passed through a low-pass filter before digitisation. Explain why this can improve the quality of earthquake records, and identify one case where an unsuitable cutoff frequency could reduce useful scientific information. [3]

2.4. **Open-ended synoptic task.** A second station at a different distance reports a larger peak voltage but smaller estimated ground displacement than the first station. Discuss how this could occur physically.

You should refer to at least **three** relevant ideas chosen from: instrument response, dominant frequency content, attenuation/absorption, local site effects, orientation/polarisation, clipping/saturation, calibration. [5]

Sample Solutions (Variant 2)

Q1

1.1

$$\omega_0 = \sqrt{\frac{k}{m}} = \sqrt{\frac{120}{0.80}} = \sqrt{150} = 12.25\,\mathrm{rad\,s^{-1}} \approx 12.2\,\mathrm{rad\,s^{-1}}.$$

$$f_0 = \frac{\omega_0}{2\pi} = \frac{12.25}{2\pi} = 1.95\,\mathrm{Hz}.$$

1.2

$$c_c = 2\sqrt{km} = 2\sqrt{120 \times 0.80} = 2\sqrt{96} = 19.6\,\mathrm{N\,s\,m^{-1}}.$$

$$\zeta = \frac{c}{c_c} = \frac{5.5}{19.6} = 0.28.$$

Since $\zeta < 1$, the system is **underdamped**.

1.3

$$\omega = 2\pi(1.80) = 11.31\,\mathrm{rad\,s^{-1}}, \quad m\omega^2 = 0.8(11.31)^2 \approx 102.3\,\mathrm{N\,m^{-1}}.$$

$$k - m\omega^2 \approx 120 - 102.3 = 17.7\,\mathrm{N\,m^{-1}}, \quad c\omega = 5.5(11.31) = 62.2\,\mathrm{N\,s\,m^{-1}}.$$

$$Y = \frac{(102.3)(1.8 \times 10^{-3})}{\sqrt{(17.7)^2 + (62.2)^2}} = \frac{0.184}{64.7} = 2.85 \times 10^{-3}\,\mathrm{m}.$$

1.4

$$L = N(2\pi r) = 900 \times 2\pi(0.012) = 67.9\,\mathrm{m}.$$

1.5

$$v_{\mathrm{rel,max}} = \omega Y = (11.31)(2.85 \times 10^{-3}) = 3.23 \times 10^{-2}\,\mathrm{m\,s^{-1}}.$$

$$\varepsilon_{\max} = BLv = (0.50)(67.9)(3.23 \times 10^{-2}) = 1.10\,\mathrm{V}.$$

1.6 (indicative points) Possible modifications include:

- Increase damping toward near-critical around operating band to reduce resonant ringing and waveform distortion; trade-off: reduced peak sensitivity.
- Lower natural frequency (larger m or softer spring) to widen flat response over 2.5 Hz to 6.0 Hz; trade-off: larger instrument, slower recovery and potential drift.
- Add electronic feedback (force-balance/semi-active damping) to linearise response and extend dynamic range; trade-off: complexity, power demand, calibration dependence.

Any two well-argued modifications earn full credit.

Q2

2.1 Time difference:

$$\Delta t = t_S - t_P = 50\,\mathrm{s}.$$

Using

$$\Delta t = d\left(\frac{1}{v_S} - \frac{1}{v_P}\right),$$

$$d = \frac{50}{\frac{1}{4.3} - \frac{1}{7.6}} \approx 494\,\mathrm{km}.$$

2.2 Each pick has $\pm 1\,\mathrm{s}$, so worst-case for Δt is $\pm 2\,\mathrm{s}$. Hence fractional uncertainty in d equals that in Δt:

$$\frac{\Delta d}{d} = \frac{2}{50} = 0.04 \Rightarrow \Delta d \approx 0.04(494) \approx 20\,\mathrm{km}.$$

So distance $\approx 494(20)\,\mathrm{km}$.

2.3 Low-pass filtering suppresses high-frequency electrical/environmental noise, improving signal-to-noise ratio and reducing aliasing before sampling. If cutoff is too low, genuine high-frequency seismic arrivals (important for onset timing or source characterisation) are attenuated, reducing scientific value.

2.4 (indicative discussion) Possible explanation: one station may receive higher-frequency waves that produce larger induced emf ($\varepsilon \propto v$), yet actual displacement is smaller because velocity can be high even when displacement amplitude is low. Local ground amplification can boost certain frequencies. Different instrument transfer functions or calibration errors can change voltage-displacement conversion. Sensor orientation relative to wave polarisation can alter measured component amplitudes. Clipping at one site can distort inferred displacement.

Mark Scheme (CIE P4 style) – Variant 2 only

Question 1 [20 marks]

Part	Marking points	Marks
1.1	Correct formula $\omega_0 = \sqrt{k/m}$; substitution; $f_0 = \omega_0/2\pi \approx 1.95\,\text{Hz}$.	3
1.2	$c_c = 2\sqrt{km}$ correct; numerical value; $\zeta = c/c_c$; correct classification (underdamped).	4
1.3	Correct use of given amplitude expression; correct ω and key terms; $Y \approx 2.85 \times 10^{-3}\,\text{m}$ (allow ecf).	4
1.4	Derivation/use $L = N(2\pi r)$; correct numerical $L \approx 67.9\,\text{m}$.	2
1.5	Use $v_{\max} = \omega Y$; use $\varepsilon = BLv$; correct $\varepsilon_{\max} \approx 1.10\,\text{V}$ (allow ecf).	3
1.6	Any two justified modifications (physics principle + impact + trade-off). 2 marks each, max 4.	4

Question 2 [15 marks]

Part	Marking points	Marks
2.1	Correct time difference; correct rearrangement; correct distance $\approx 494\,\text{km}$.	3
2.2	Correct worst-case Δt uncertainty; propagation to distance; final uncertainty statement (about $\pm 20\,\text{km}$).	4
2.3	Clear reason for low-pass benefit (noise/aliasing); valid risk from low cutoff removing useful seismic content.	3
2.4	Any three relevant, physically correct discussion points (1 mark each) + coherent comparison/interpretation up to 2 marks.	5

Total: 35 marks